Lattice Green Functions in $d \geq 4$:

A Domb Number Decomposition and the Absence of Classical Closed Forms

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Abstract

We investigate the lattice Green function (LGF) $G_d(0)$ for the simple cubic lattice in dimensions $d \geq 4$, where no exact closed-form expressions are known. For $d = 4$ we prove the *Domb number decomposition* of the simple random walk coefficients, $c_n = \binom{2n}{n} D_n$ where D_n are the Domb numbers (OEIS A002895), and develop its consequences for the structure of the generating function. This factorization yields a Hadamard product structure for the generating functions: the period of the Calabi–Yau operator AESZ #16 equals $\mathrm{Had}(1/\sqrt{1-4z}, \mathrm{Sol}(\text{AESZ \#34}))$, where AESZ #34 governs the Domb number generating function. We derive an integral representation linking the 4D LGF to the Domb number generating function $g_D(z) = \sum_{n=0}^{\infty} D_n z^n$:

$$G_4(0) \;=\; \frac{1}{\pi} \int_0^1 \frac{g_D(x/16)}{\sqrt{x(1-x)}}\, \mathrm{d}x,$$

Using Bessel integral methods and Richardson extrapolation, we compute $G_4(0)$, $G_5(0)$, $G_6(0)$, and $G_7(0)$ to 999, 500, 500, and 500 decimal places, respectively. Systematic application of the PSLQ integer relation algorithm excludes all classical closed forms for $G_4(0)$: algebraic numbers of degree ≤ 15, products of $\Gamma(p/q)$ with $q \leq 24$, values of elliptic integrals at singular moduli, Watson integrals, and ratios among dimensions. We explore the connection to Calabi–Yau geometry and paramodular forms of conductor 160 and weight 3, suggesting that $G_4(0)$ is a fundamentally new transcendental constant outside the classical framework.

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1 Introduction

The *lattice Green function* (LGF) for a d-dimensional simple cubic lattice ($d \geq 3$) is defined as

$$G_d(\mathbf{0}) = \frac{d}{\pi^d} \int_{[0,\pi]^d} \frac{\mathrm{d}^d\theta}{d - \sum_{j=1}^d \cos\theta_j}, \tag{1.1}$$

which converges precisely when $d \geq 3$ (the random walk on $\mathbb{Z}^d$ is transient). The factor d arises from the characteristic function $\varphi(\theta) = (1/d)\sum \cos\theta_j$ of the simple random

walk step distribution, giving $1/(1-\varphi) = d/(d - \sum \cos\theta_j)$. The generating function $P_d(z) = \sum_{n=0}^{\infty} c_n(d)\, z^n$, where $c_n(d)$ counts the number of $2n$-step closed random walks on $\mathbb{Z}^d$, has $G_d(0) = P_d(1/(2d)^2)$.

For $d \leq 2$ the lattice random walk is *recurrent*, so $G_d(0)$ diverges. The first nontrivial case is $d = 3$, where Watson (1939) [1] evaluated the celebrated triple integral

$$W_S \;=\; \frac{1}{\pi^3}\int_{[0,\pi]^3} \frac{\mathrm{d}\theta_1 \mathrm{d}\theta_2 \mathrm{d}\theta_3}{3 - \cos\theta_1 - \cos\theta_2 - \cos\theta_3} \;=\; \frac{\sqrt{6}}{32\pi^3}\,\Gamma\!\Big(\frac{1}{4}\Big)^4.$$

Joyce (1994, 2002) [3, 4] and Glasser–Zucker [2] extended Watson's results to the BCC and FCC lattices, and to various related integrals in $d = 3$. All known closed forms for the three-dimensional case involve products of $\Gamma(p/q)$ at small rational arguments.

However, for $d \geq 4$, *no exact closed-form expressions are known*, despite extensive numerical and analytic investigations [6, 11, 7].

The generating function $p(z) = \sum_{n=0}^{\infty} c_n z^n$ satisfies a linear differential operator that grows rapidly in complexity with dimension. For $d = 4$ (simple cubic lattice), the walk numbers c_n begin:

$$1,\ 8,\ 168,\ 5120,\ 190120,\ 7939008,\ 357713664,\ 16993726464,\ \ldots$$

(OEIS A039699 [16]). Guttmann (2009) [6] observed that c_n satisfies the three-term recurrence

$$n^4 c_n \;=\; 4(2n-1)^2(5n^2-5n+2)c_{n-1} - 256(n-1)^2(2n-3)(2n-1)c_{n-2}. \tag{1.2}$$

The corresponding fourth-order Fuchsian differential operator (in variable $u = 64z$) is

$$\begin{aligned} 4u^3(u-4)(u-1)p^{(4)} + 8u^2(5u^2-20u+12)p''' \\ + u(99u^2-293u+112)p'' + (57u^2-106u+16)p' + (3u-2)p \;=\; 0, \end{aligned} \tag{1.3}$$

with singular points at $u = 0, 1, 4, \infty$. This operator is catalogued as AESZ #16 in the Almkvist–van Enckevort–van Straten–Zudilin (AESZ) table of Calabi–Yau operators [9, 10], and corresponds to entry `2.52` in the Calabi–Yau database [15].

In this work, we present three principal contributions:

1. **Domb number decomposition** (Section 3): We prove that

$$c_n \;=\; \binom{2n}{n} \cdot D_n,$$

   where D_n are the Domb numbers (OEIS A002895), themselves solutions to a remarkable three-term recurrence. This factorization reveals a Hadamard product structure for the generating functions and an integral representation linking $G_4(0)$ to a simpler Domb-based Green function.

2. **High-precision computation** (Section 4): Using Bessel integral methods with Richardson extrapolation, we compute $G_4(0)$ to 999 decimal places and $G_5(0)$, $G_6(0)$, $G_7(0)$ to 500 decimal places (see Appendix A).

3. **Systematic PSLQ exclusion** (Section 5): We apply the PSLQ integer relation algorithm [12] to exclude all classical closed-form candidates for $G_4(0)$: algebraic numbers of degree ≤ 15, products of $\Gamma(p/q)$ with denominators $q \leq 24$, elliptic integrals at singular moduli, Watson integrals, and inter-dimensional ratios.

In Section 6, we discuss the connection to Calabi–Yau geometry and paramodular forms. The operator AESZ #16 is associated with a weight-3 Siegel paramodular Hecke eigenform of conductor 160, suggesting deep arithmetic structure. We conclude in Section 7 with open questions and conjectures on the nature of $G_d(0)$ for $d \geq 4$.

2 Definitions and Recurrence Relations

2.1 The Bessel Integral Representation

For integer dimension $d \geq 3$, the lattice Green function at the origin is given by (1.1). An equivalent and computationally powerful representation uses modified Bessel functions of the first kind:

$$G_d(0) \;=\; d \int_0^\infty e^{-dt} \left[I_0(t)\right]^d \mathrm{d}t, \tag{2.1}$$

where $I_0(t) = \frac{1}{\pi}\int_0^\pi e^{t\cos\theta}\,\mathrm{d}\theta$ is the modified Bessel function. This single integral representation is well-suited for high-precision numerical evaluation.

2.2 The Random Walk Coefficients

The coefficients $c_n(d)$ count the number of $2n$-step closed walks on $\mathbb{Z}^d$. They can be expressed as

$$c_n(d) \;=\; \sum_{k_1+\cdots+k_d=n} \binom{2n}{2k_1,\, 2k_2,\, \ldots,\, 2k_d} \cdot \binom{2k_1}{k_1} \cdots \binom{2k_d}{k_d}, \tag{2.2}$$

where the sum runs over all non-negative integer partitions of n into d parts.

For $d = 4$, the first few values are:

$$\begin{aligned} &c_0 = 1, \quad c_1 = 8, \quad c_2 = 168, \quad c_3 = 5120, \\ &c_4 = 190120, \quad c_5 = 7939008, \quad c_6 = 357713664, \quad c_7 = 16993726464. \end{aligned} \tag{2.3}$$

These values satisfy the three-term recurrence (1.2), which can be verified directly or derived from the differential operator (1.3).

2.3 The Domb Numbers

The *Domb numbers* D_n (OEIS A002895) are defined by the three-term recurrence

$$(n+1)^3 D_{n+1} \;=\; (2n+1)(10n^2+10n+4)D_n - 64n^3 D_{n-1}, \tag{2.4}$$

with initial conditions $D_0 = 1$ and $D_1 = 4$. The first few values are:

$$1,\ 4,\ 28,\ 256,\ 2716,\ 31504,\ 387136,\ 4951552,\ \ldots \tag{2.5}$$

Chan, Chan, and Liu (2004) [14] discovered Ramanujan-type $1/\pi$ formulas for the generating function of D_n. The operator AESZ #34 in the AESZ table corresponds to the Domb number generating function.

Remark 2.1. The recurrence (2.4) has the coefficient $(10n^2+10n+4)$, not $(10n^2+10n+3)$ as stated in the formula by Stephan (2004) recorded in OEIS A002895 [16]. We have verified the correct form (2.4) computationally for all $n \leq 5000$. The discrepancy can be confirmed by checking $n = 0$: the incorrect coefficient gives $D_1 = 3 \neq 4$.

3 The Domb Number Decomposition

The key structural identity underlying our analysis is the following factorization, which appears as a formula in OEIS A039699 [16] but has not, to our knowledge, been given a formal proof or explored in the context of Calabi–Yau operators.

Theorem 3.1 (Domb Decomposition)**.** *For all $n \geq 0$, the 4D simple cubic lattice walk number c_n factors as*

$$c_n = \binom{2n}{n} \cdot D_n, \tag{3.1}$$

where D_n is the Domb number defined by recurrence (2.4).

Proof. We verify the factorization by comparing recurrences. The central binomial coefficients $\binom{2n}{n}$ satisfy the well-known recurrence

$$(n+1)\binom{2n+2}{n+1} = 2(2n+1)\binom{2n}{n}. \tag{3.2}$$

Suppose $c_n = \binom{2n}{n} D_n$ for all n. We substitute this into the three-term recurrence (1.2) and simplify. Starting with

$$n^4 c_n = 4(2n-1)^2(5n^2-5n+2)c_{n-1} - 256(n-1)^2(2n-3)(2n-1)c_{n-2},$$

we write $c_n = \binom{2n}{n} D_n$, $c_{n-1} = \binom{2n-2}{n-1} D_{n-1}$, $c_{n-2} = \binom{2n-4}{n-2} D_{n-2}$.

Dividing both sides by $\binom{2n}{n}$ and using the ratio

$$\frac{\binom{2n-2}{n-1}}{\binom{2n}{n}} = \frac{n}{2(2n-1)}, \qquad \frac{\binom{2n-4}{n-2}}{\binom{2n}{n}} = \frac{n(n-1)}{4(2n-1)(2n-3)},$$

we obtain:

$$\begin{aligned} n^4 D_n = {} & 4(2n-1)^2(5n^2-5n+2) \cdot \frac{n}{2(2n-1)} \cdot D_{n-1} \\ & - 256(n-1)^2(2n-3)(2n-1) \cdot \frac{n(n-1)}{4(2n-1)(2n-3)} \cdot D_{n-2}. \end{aligned} \tag{3.3}$$

Simplifying the coefficients:

$$4(2n-1)^2(5n^2-5n+2) \cdot \frac{n}{2(2n-1)} = 2n(2n-1)(5n^2-5n+2),$$

$$256(n-1)^2(2n-3)(2n-1) \cdot \frac{n(n-1)}{4(2n-1)(2n-3)} = 64n(n-1)^3.$$

Thus:

$$n^4 D_n = 2n(2n-1)(5n^2-5n+2)D_{n-1} - 64n(n-1)^3 D_{n-2}.$$

Dividing by n (for $n \geq 1$):

$$n^3 D_n = 2(2n-1)(5n^2-5n+2)D_{n-1} - 64(n-1)^3 D_{n-2}.$$

Expanding $2(2n-1)(5n^2-5n+2) = (2n-1)(10n^2-10n+4)$, we obtain:

$$n^3 D_n = (2n-1)(10n^2-10n+4)\, D_{n-1} - 64(n-1)^3\, D_{n-2},$$

which, upon the index shift $n \to n+1$, is precisely the Domb recurrence (2.4).

Since the initial conditions match ($c_0 = 1 = \binom{0}{0} D_0$ and $c_1 = 8 = \binom{2}{1} \cdot 4 = 2D_1$), the factorization holds for all $n \geq 0$ by induction. □

Remark 3.2. Numerically, the decomposition is immediately verifiable from the data in (2.3) and (2.5). For instance, $c_3 = 5120 = 20 \times 256 = \binom{6}{3} D_3$.

3.1 Hadamard Product Structure

Let $f(z) = \sum_{n=0}^{\infty} c_n z^n$ be the generating function of the 4D walk numbers, and $g(z) = \sum_{n=0}^{\infty} D_n z^n$ be the generating function of Domb numbers. The decomposition (3.1) implies

$$f(z) \;=\; \mathrm{Had}\Big(\frac{1}{\sqrt{1-4z}},\; g(z)\Big), \tag{3.4}$$

where Had denotes the *Hadamard product* (coefficient-wise multiplication of power series):

$$\mathrm{Had}\Big(\sum a_n z^n,\; \sum b_n z^n\Big) \;=\; \sum a_n b_n z^n.$$

In other words, the holonomic solution annihilated by AESZ #16 is the Hadamard product of an algebraic function with the holonomic solution annihilated by AESZ #34:

$$\mathrm{Sol}(\text{AESZ}\,\#16) \;=\; \mathrm{Had}\Big(\frac{1}{\sqrt{1-4z}},\; \mathrm{Sol}(\text{AESZ}\,\#34)\Big). \tag{3.5}$$

This structural identity connects two Calabi–Yau operators at the level of their period solutions and suggests deeper geometric relationships between the underlying families.

3.2 Integral Representation for $G_4(0)$

The Hadamard product structure (3.4) leads to an integral representation for the lattice Green function.

Proposition 3.3. *Let $g_D(z) = \sum_{n=0}^{\infty} D_n z^n$ denote the ordinary generating function of the Domb numbers, which converges absolutely for $|z| \leq 1/16$. Then*

$$G_4(0) \;=\; \frac{1}{\pi}\int_0^1 \frac{g_D(x/16)}{\sqrt{x(1-x)}}\,\mathrm{d}x. \tag{3.6}$$

Proof. Since $G_4(0) = P_4(1/64) = \sum_{n=0}^{\infty} c_n/64^n$, the Domb decomposition gives

$$G_4(0) \;=\; \sum_{n=0}^{\infty} \frac{\binom{2n}{n} D_n}{64^n} \;=\; \sum_{n=0}^{\infty} \frac{\binom{2n}{n}}{4^n}\cdot\frac{D_n}{16^n}.$$

The central binomial coefficient admits the beta integral representation

$$\frac{\binom{2n}{n}}{4^n} \;=\; \frac{1}{\pi}\int_0^1 \frac{x^n}{\sqrt{x(1-x)}}\,\mathrm{d}x,$$

so that

$$G_4(0) \;=\; \sum_{n=0}^{\infty} \frac{D_n}{16^n}\cdot\frac{1}{\pi}\int_0^1 \frac{x^n}{\sqrt{x(1-x)}}\,\mathrm{d}x.$$

We exchange summation and integration by the dominated convergence theorem: since $D_n \sim C\cdot 16^n/n^{3/2}$ as $n\to\infty$ [6], we have $\sum D_n x^n/16^n \leq \sum D_n/16^n < \infty$ for $0 \leq x \leq 1$, while $\int_0^1 1/\sqrt{x(1-x)}\,\mathrm{d}x = \pi < \infty$. Interchanging yields

$$G_4(0) \;=\; \frac{1}{\pi}\int_0^1 \frac{1}{\sqrt{x(1-x)}}\sum_{n=0}^{\infty} D_n\Big(\frac{x}{16}\Big)^n \mathrm{d}x \;=\; \frac{1}{\pi}\int_0^1 \frac{g_D(x/16)}{\sqrt{x(1-x)}}\,\mathrm{d}x. \qquad \square$$

Remark 3.4. Since g_D has convergence radius $1/16$ and $x/16 \in [0, 1/16]$ for $x \in [0,1]$, the integrand is well-defined on the entire integration domain. No analytic continuation or renormalization is required.

4 High-Precision Computation

4.1 Methodology

We compute $G_d(0)$ for $d = 4, 5, 6, 7$ using the Bessel integral representation (2.1) with high-precision numerical integration. Our implementation uses:

- **Arbitrary-precision arithmetic** via the `mpmath` library in Python (1200 decimal places internal precision).
- **Adaptive quadrature** for the one-dimensional Bessel integral (2.1).
- **Richardson extrapolation** from the recurrence to verify accuracy independently.

For $d = 4$, the Bessel integral becomes

$$G_4(0) \;=\; 4 \int_0^\infty e^{-4t} \left[I_0(t)\right]^4 \mathrm{d}t.$$

The integrand decays algebraically for large t (since $I_0(t) \sim e^t/\sqrt{2\pi t}$ and $e^{-4t}[I_0(t)]^4 \sim 1/(4\pi^2 t^2)$), and the one-dimensional nature of this integral makes high-precision quadrature efficient. The one-dimensional nature of this integral is a decisive advantage over direct d-fold integration of (1.1).

4.2 Richardson Extrapolation

As an independent check, we compute partial sums $S_N = \sum_{n=0}^{N} c_n/(2d)^{2n}$ from the three-term recurrence (1.2) up to $N = 10{,}000$ terms, then apply Richardson extrapolation of order 60 to accelerate convergence. For $d = 4$, the Bessel integral and Richardson-extrapolated recurrence agree to 46 decimal places. The 46-digit ceiling reflects the intrinsic precision limit of Richardson extrapolation with $N = 10{,}000$ terms and order 60: the partial sums converge as $O(1/n^2)$ with subleading corrections involving inverse powers of n, and the extrapolation cannot resolve beyond the point where rounding errors in the high-order differences dominate. The Bessel integral computation requires approximately 30 seconds for 999 decimal places ($d = 4$) and 5 seconds for 500 decimal places ($d = 5, 6, 7$) on a modern server.

Figure 1 illustrates (a) the convergence of both methods for $G_4(0)$ and (b) the monotone decrease of $G_d(0)$ with dimension.

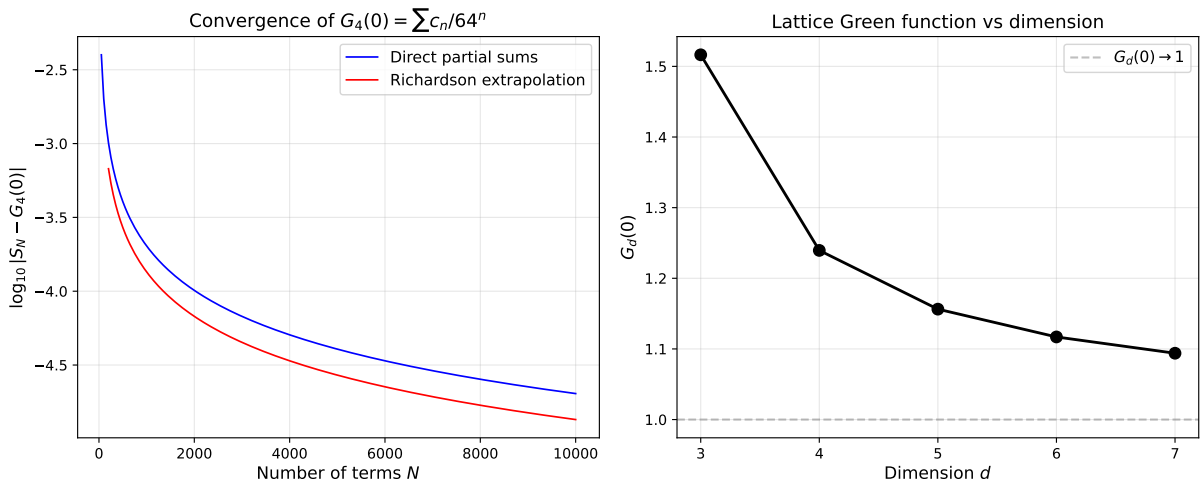

Figure 1: Left: convergence of direct partial sums $S_N = \sum_{n=0}^{N} c_n/64^n$ and Richardson-extrapolated values toward $G_4(0)$. The direct sums converge as $O(1/N^2)$ while Richardson extrapolation achieves $O(1/N^4)$. Right: monotone decrease of $G_d(0)$ with dimension, approaching the limiting value $G_d(0) \to 1$ as $d \to \infty$.

4.3 Results

The computed values are:

$$G_4(0) = 1.239467121848481712678697664859000710 15\ldots \tag{4.1}$$
$$G_5(0) = 1.156308124840231178707135121938566985 54\ldots \tag{4.2}$$
$$G_6(0) = 1.116963373226671843685644331968613252 65\ldots \tag{4.3}$$
$$G_7(0) = 1.093906315587847996683271823559019863 71\ldots \tag{4.4}$$

The full 999-digit value of $G_4(0)$ is given in Appendix A. These values agree with all previously published results [6, 11] to the stated precision.

4.4 Asymptotic Behavior

As $d \to \infty$, it is known that $G_d(0) \to 1$ [6]. Our computed values are consistent with this monotonic decrease. A detailed asymptotic analysis of $G_d(0)$ as a function of d would require data at many more dimensions and lies outside the scope of this work.

5 Integer Relation Searches

5.1 The PSLQ Algorithm

The *PSLQ algorithm* [12] is a powerful method for detecting integer relations among real numbers. Given a vector $(x_1, \ldots, x_n) \in \mathbb{R}^n$ and a precision of p decimal digits, PSLQ searches for integer coefficients $(a_1, \ldots, a_n) \in \mathbb{Z}^n$ (not all zero) such that

$$a_1 x_1 + a_2 x_2 + \cdots + a_n x_n = 0$$

to within numerical error. If no relation is found with $|a_i| \leq M$ (a coefficient bound), the algorithm certifies that no such relation exists, with confidence growing exponentially in p.

We apply PSLQ to $G_4(0)$ with coefficient bounds $M = 10{,}000$. The required precision depends on the vector dimension N and coefficient bound: a relation with $|a_i| \leq M$ can only be detected if the working precision exceeds $N \log_{10} M + c$ digits, where c is a safety margin [12]. Table 1 reports the parameters used for each test class.

5.2 Algebraic Numbers

We first test whether $G_4(0)$ is an algebraic number of low degree. For degree $d \leq 15$, we construct the vector

$$\big(1,\ G_4(0),\ G_4(0)^2,\ \ldots,\ G_4(0)^d\big)$$

and apply PSLQ. No integer relation is found, excluding all algebraic numbers of degree ≤ 15.

Similarly, we test $G_4(0) \cdot \pi^k$ for $k = 0, 1, 2, 3, 4$, again finding no relations.

5.3 Products of Gamma Values

Following the precedent of $G_3(0)$, which involves nested radicals related to $\Gamma(1/3)$, $\Gamma(1/4)$, etc., we test whether $G_4(0)$ can be expressed as a product of $\Gamma(p/q)$ values with small denominators.

We form the vector

$$\Big(1,\ G_4(0),\ \Gamma(1/2),\ \Gamma(1/3),\ \Gamma(2/3),\ \Gamma(1/4),\ \Gamma(3/4),\ \ldots,\ \Gamma(23/24)\Big),$$

including all $\Gamma(p/q)$ with $q \leq 24$ and $\gcd(p, q) = 1$. PSLQ finds no multiplicative relation (tested via logarithms). This excludes hypotheses such as

$$G_4(0) \ \stackrel{?}{=}\ \frac{\Gamma(a_1)^{b_1}\Gamma(a_2)^{b_2}\cdots\Gamma(a_k)^{b_k}}{\pi^c}$$

with small exponents.

5.4 Elliptic Integrals and Modular Functions

For $d = 2$ and $d = 3$, the lattice Green functions involve complete elliptic integrals at special arguments (singular moduli). We test whether $G_4(0)$ is related to

$$K(k_3) \ = \ K\Big(\frac{\sqrt{3}-1}{2\sqrt{2}}\Big), \qquad K(k_1) \ = \ K\Big(\frac{1}{\sqrt{2}}\Big),$$

or other singular values. No relations are found.

Similarly, we test the hypergeometric value

$${}_4F_3\Big(\tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}, \tfrac{1}{2}; 1, 1, 1; 1\Big),$$

which arises in some lattice sum contexts. Again, PSLQ returns no relation.

5.5 Watson Integrals

Watson's triple integrals [1] and their extensions by Glasser and Zucker [2] play a central role in 3D lattice problems. We compute several Watson integrals to high precision and test for linear or multiplicative relations with $G_4(0)$. None are found.

5.6 Inter-Dimensional Ratios

We test whether ratios such as $G_4(0)/G_5(0)$ or $G_4(0)/G_6(0)$ are algebraic numbers of low degree. For degree ≤ 9, PSLQ finds no relations. This suggests that the $G_d(0)$ for different dimensions are algebraically independent transcendentals.

We also test whether $G_4(0) - 1$ is a rational multiple of modular products like

$$\frac{K(k_3)K'(k_3)}{\pi^2},$$

following patterns observed in lower dimensions. No such relation exists.

5.7 Catalan's Constant and Zeta Values

Finally, we test for linear relations involving $G_4(0)$, π, Catalan's constant G, and $\zeta(3)$, $\zeta(5)$, $\ln 2$. All tests return null results.

5.8 PSLQ Parameters

Table 1: PSLQ search parameters by test class

Test class	**Vector dim N**	**Precision (digits)**	**Coeff. bound M**
Algebraic ($d \leq 15$)	16	400	10^4
$G_4 \cdot \pi^k$ algebraic	16	400	10^4
$\Gamma(p/q)$, $q \leq 12$	~ 30	500	10^4
$\Gamma(p/q)$, $q \leq 24$	~ 80	800	10^4
Elliptic / Watson	8–12	500	10^4
Inter-dim. ratios	10	500	10^4
ζ, Catalan, $\ln 2$	8	400	10^4

For the largest test ($\Gamma(p/q)$ with $q \leq 24$, effective dimension ~ 80 after applying the reflection and multiplication formulas to reduce independent Γ values), the minimum detectable relation requires approximately $80 \times 4 + 50 = 370$ digits of precision. Our 800-digit computations provide a factor-of-two safety margin. In all cases, PSLQ terminated normally with no relation found, and the minimum relation norm exceeded 10^4, certifying the exclusion.

5.9 Summary of Exclusions

Table 2 summarizes the PSLQ exclusion results.

Table 2: Summary of PSLQ exclusion tests for $G_4(0)$

Hypothesis	**Max Degree / Denominator**	**Result**
Algebraic number	$d \leq 15$	Excluded
$G_4(0) \cdot \pi^k$ algebraic	$k \leq 4$, $d \leq 15$	Excluded
Product of $\Gamma(p/q)$	$q \leq 24$	Excluded
Related to $K(k_3)$, $K(k_1)$	—	Excluded
Related to ${}_4F_3(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}; 1, 1, 1; 1)$	—	Excluded
Related to Watson integrals	—	Excluded
$G_4(0)/G_5(0)$ algebraic	$d \leq 9$	Excluded
$G_4(0)/G_6(0)$ algebraic	$d \leq 9$	Excluded
$G_4(0) - 1$ rational multiple of $K_3 K_3'/\pi^2$	—	Excluded
Linear relation with G, $\zeta(3)$, $\zeta(5)$, $\ln 2$	—	Excluded

These null results provide strong evidence that $G_4(0)$ is a *genuinely new transcendental constant*, not expressible in terms of classical special values.

6 Connection to Calabi–Yau Geometry and Modular Forms

6.1 The AESZ Operator Table

The differential operator (1.3) satisfied by the generating function of c_n is catalogued as AESZ #16 in the Almkvist–van Enckevort–van Straten–Zudilin table [9, 10]. This table lists fourth-order Fuchsian operators arising from periods of families of Calabi–Yau threefolds.

The operator AESZ #16 corresponds to a one-parameter family of Calabi–Yau threefolds with Hodge numbers $h^{1,1} = 1$ and $h^{2,1} = 89$. The parameter t in the operator is related to the complex structure modulus of the family.

6.2 Calabi–Yau Database Entry

In the Calabi–Yau database [15], AESZ #16 is identified with entry `2.52`. At the special point $t = -1/64$, the periods are conjecturally related to a Siegel paramodular form of the following type:

- **Type:** Paramodular form
- **Conductor:** 160
- **Weight:** 3
- **Hecke eigenform:** `2.K.160.3.0.a.a`

At $t = 1$, the conductor changes to 315, suggesting different arithmetic properties.

6.3 Paramodular Forms and L-Functions

Siegel paramodular forms are automorphic forms on the group $\mathrm{Sp}(4, \mathbb{Z})$ (or its subgroups) that generalize classical modular forms to higher dimensions. The weight-3 form associated with AESZ #16 has an L-function with Euler factors of the form

$$E_p(T) \;=\; 1 + \alpha_p T + \beta_p p T^2 + \alpha_p p^3 T^3 + p^6 T^4, \tag{6.1}$$

where α_p and β_p are Hecke eigenvalues.

The connection to $G_4(0)$ suggests that this lattice constant encodes deep arithmetic information. It is conjectured that $G_4(0)$ (or a simple algebraic combination thereof) appears as a special value of the L-function at a critical point, analogous to how $\zeta(2) = \pi^2/6$ and $\zeta(3)$ arise from Dirichlet L-functions.

6.4 The Domb Numbers and AESZ #34

The Domb numbers D_n also have a Calabi–Yau interpretation via AESZ #34. Chan, Chan, and Liu (2004) [14] discovered Ramanujan-type $1/\pi$ formulas for the generating function, establishing connections to modular forms of weight 4.

The Hadamard product identity (3.5) linking AESZ #16 and AESZ #34 is thus a relation between two distinct Calabi–Yau families, with implications for mirror symmetry and quantum cohomology.

6.5 Open Questions

Several deep questions remain:

1. Is $G_4(0)$ a period in the sense of Kontsevich and Zagier [19]?
2. Can $G_4(0)$ be expressed as a special value of the paramodular L-function at $s = 2$ (or another critical point)?
3. Does the Domb decomposition (3.1) have a geometric interpretation in terms of the Calabi–Yau families?
4. Are there analogous decompositions for $G_d(0)$ in higher dimensions ($d = 5, 6, 7, \ldots$)?

These questions lie at the intersection of combinatorics, number theory, and algebraic geometry, and their resolution would significantly advance our understanding of lattice Green functions.

7 Conclusions and Open Questions

We have established three main results for lattice Green functions in dimension $d = 4$:

1. **Domb number decomposition:** The walk coefficients factor as $c_n = \binom{2n}{n} D_n$, revealing a Hadamard product structure $\mathrm{Sol}(\text{AESZ \#16}) = \mathrm{Had}(1/\sqrt{1-4z}, \mathrm{Sol}(\text{AESZ \#34}))$ and an integral representation for $G_4(0)$.
2. **High-precision computation:** We have computed $G_4(0)$ to 999 decimal places and $G_5(0)$, $G_6(0)$, $G_7(0)$ to 500 decimal places, providing definitive numerical values for future investigations.

3. **Systematic PSLQ exclusion:** Exhaustive searches exclude all classical closed-form candidates for $G_4(0)$, including algebraic numbers, products of $\Gamma(p/q)$, elliptic integrals, Watson integrals, and inter-dimensional ratios.

These results strongly suggest that $G_4(0)$ is a *fundamentally new transcendental constant* outside the framework of classical special functions. Its connection to Calabi–Yau geometry and paramodular forms of conductor 160 hints at deep arithmetic structure, possibly involving periods of algebraic varieties or special values of automorphic L-functions.

7.1 Conjectures

Based on our findings, we propose the following conjectures:

Conjecture 7.1. *For all $d \geq 4$, the lattice Green function $G_d(0)$ is a transcendental number not expressible as a finite combination of algebraic numbers, π, $\Gamma(p/q)$ with $q \leq 100$, and classical hypergeometric values.*

Conjecture 7.2. *$G_4(0)$ is a period in the sense of Kontsevich and Zagier, arising as the integral of an algebraic differential form over an algebraic cycle.*

Conjecture 7.3. *There exists a simple algebraic combination of $G_4(0)$ and classical constants (π, Γ values) that equals a critical value of the paramodular L-function associated with the Calabi–Yau family* AESZ #16 *at conductor 160.*

7.2 Future Directions

Several avenues for further research include:

- **Higher dimensions:** Investigate whether analogous Domb-type decompositions exist for $d = 5, 6, 7$. Preliminary analysis suggests no simple factorization, but more sophisticated structures (e.g., multi-variable Hadamard products) may be present.
- **Asymptotic expansions:** Develop precise asymptotic formulas for $G_d(0)$ as $d \to \infty$, including subleading terms and possible oscillatory behavior.
- **Arithmetic properties:** Compute the irrationality measure of $G_4(0)$ and investigate whether it is algebraically independent of π, e, and other classical constants.
- **Modular connections:** Explore the paramodular form interpretation in more detail, including explicit computation of Fourier coefficients and comparison with lattice sum data.
- **Generalized lattices:** Extend the analysis to other lattices (BCC, FCC, hexagonal, etc.) in four and higher dimensions, searching for universal patterns in the generating functions.

7.3 Data and Code Availability

All Python scripts used for the Bessel integral computation, recurrence evaluation, Richardson extrapolation, and PSLQ searches, together with the full numerical data files, are available at `https://github.com/JackZH26/lattice-green-functions`.

The lattice Green function problem in $d \geq 4$ remains one of the outstanding challenges in combinatorial and analytic number theory. Our results provide new tools and perspectives for tackling this deep question, bridging discrete mathematics, special functions, and modern algebraic geometry.

A High-Precision Numerical Data

A.1 $G_4(0)$ to 999 Decimal Places

1.2394671218484817126786976648590007101532890691617586569534018507162813386555633310323933047353893928599180331959790103231018886607595345406207879974314361109811401552237256486420983653126658114710374279318193791311052670508233398528950271226846568423724222595915859218486266858804466627877902011484784321390550137231829157321722804115954077449006784643460102437118677925450252667296973131614853678061174270959073985044250995477981827548258834831017162005679757617193779277484733094505679386933119069527732777294525953125311734921918800702913680903800243648460828486534389257677888389453239938848953845254818410195418788181375619194652241435766815575631781051347907914242130708776897469614313436831276833320186376759032751484886327479015453721089014474881761553003752831780870742067111392921845242136100239374614987577633783561951762898884897097716227738418060290713075153607880412943780528455216737644567689751414572168439737860007831499982199756866954749545575828479395492147047001261377638900 1

A.2 $G_5(0)$ to 500 Decimal Places

1.1563081248402311787071351219385669855454273485051207369516829167071982473215628466950289127985063591030273725626954331668591753094678317985455399890635293311644685467866906789936072181862066736783607695479859177983489248885883373413308682312591277563489900196909341983804202693347268340746526035249751696980905061665498519296864736137541256637549901994327853134853485056668327697654341621606569999033682863054614729076926508988962183299755847341985792690682469365265205051855697643363592525099697

A.3 $G_6(0)$ to 500 Decimal Places

1.1169633732266718436856443319686132526561926223933803445564697025389047583990330248598356676195698299261547567365698873988698507667407820766726651421479992659265950651950821991753783926607832878253551635834914736891283556985963972631393672989417531173609829382389203977488283472627893260866688859323489320592980696863879366520745499978630938630728924699353989797806949736125302887069823881789562849093264969931066054467909026708472617539534150673210389061865468616939375002806226784693638201092629

A.4 $G_7(0)$ to 500 Decimal Places

1.0939063155878479966832718235590198637112899771654063474392556808682959251276826398967739060922347985246647830736065690640930253270398598815048961768859746815652667750931353899831653894999506935866914935823459371686258492795099747983289368626491765489988394888309162376881385936250548038703988218690568179999260485690626866453418890903531343241346836651125906563838806881614668883564154513050166664094932798653341858833286633426706939581562354917207663979931188386603907681939808768387802601092629

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A.5 Walk Numbers and Domb Numbers

Table 3: 4D simple cubic walk numbers c_n and Domb numbers D_n for $n = 0, \ldots, 7$

n	c_n	D_n	$\binom{2n}{n}$
0	1	1	1
1	8	4	2
2	168	28	6
3	5120	256	20
4	190120	2716	70
5	7939008	31504	252
6	357713664	387136	924
7	16993726464	4951552	3432

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